

Vadim HEMZELLEC-DAVIDSON

 LinkedIn |  GitHub |  vadim.hemzellec-davidson@polytechnique.edu |  +33 7 87 23 61 49 |  vemz.github.io  Hugging Face

EDUCATION

École Polytechnique

2025 - 2026

Master of Science in Data Science

Relevant coursework: Advanced learning for text and graph data (ALTEGRAD - MVA), Deep Learning, Machine Learning, Optimization, Reinforcement Learning, Representation Learning for Computer Vision and Medical Imaging (MVA), Diffusion Models.

Télécom SudParis

2023 - 2026

Engineering degree in Applied Mathematics

Relevant coursework: Machine Learning, Continuous Optimization for ML, Stochastic Processes, Bayesian Inference (HMM, Particle Filtering), Applied Statistics.

Lycée Fabert

2021 - 2023

Intensive Preparatory Classes (MPSI-MP*), competitive entrance preparation for top engineering schools.

PROJECTS

Coordination Ceilings and Deferral Dynamics in Masked Diffusion LMs — Ongoing

- Formulating a deferral-based decoding strategy, committing the top token while deferring remaining positions to mitigate factorization error, in collaboration with Transformer Lab.

OVHai LLM (formerly DragonLLM) — Capstone Project M2 Data Science, 2026

- Created a token-level labeled toxicity dataset of 24K samples of toxic and safe Q&A, leveraging Qwen3-30B models.
- Developed a real-time toxicity and safety classifier to interrupt harmful generations during inference.
- Inspired by Qwen-Guardrails and circuit-breaker architectures for controllable LLM deployment.

Research Paper Studies & Re-implementations, 2025 - 2026

- Re-implemented RLOO with custom baseline and comparison with PPO / GRPO. 
- Studied and reproduced core components of CANDI: hybrid continuous/discrete diffusion models on the Code-Parrot dataset. 

Hi!ckathon (Hi!Paris) Finalist, 2025

- Finalist and ranked 2nd/68 (tied) in model performance within a team of 6 students.
- Engineered a gated model architecture aggregating a probabilistic router (XGBoost) and a regressor expert to handle zero-inflated target distributions.

Molecular Graph Captioning — ALTEGRAD Competition (MVA), 2025

- Retrieval-based graph machine learning approach to generate natural language captions from molecular graphs.
- Leveraged graph embeddings for molecule-text alignment in a chemistry-informed setting.

WORK EXPERIENCE

Airbus Helicopters, Donauwörth - AI R&D Internship

Mar 2026 - Jul 2026

- Designed and implemented an end-to-end pipeline, from raw dataset curation to model training, to fine-tune an AI model for domain-specific XML code generation, within a highly constrained defense environment.
- Curated a highly specialized dataset based on rigorous military specifications to automate functional validation scripts for NH90 helicopter components.
- Implemented the ACE (Agentic Context Engineering) framework to industrialize the solution.

Blockchain AkademIA, Metz - Internship

Jun 2025 - Aug 2025

Contributed to the regional democratization of blockchain and AI by supporting the integration of a new solution within the company and leveraging N8N for automation workflows.

Aerow, Paris - Internship

Jun 2024 - Jul 2024

Conducted an inventory of potential Research Tax Credit (CIR) eligible topics across all areas of expertise at the company.

SKILLS AND INTERESTS

Prog. Languages Python, Java, SQL, C, Bash, Git

Frameworks Pandas, Scikit-learn, Transformers, PEFT, TRL, vLLM, Matplotlib, Numpy, PyTorch

Soft skills Leadership (Former VP, Sports Committee), Teamwork (student associations)

Languages French (native), English (C1), Spanish (B2)

Interests AI, Sports (Tennis), Music (Piano 10 yrs), Audio/Video production